APM5557: Partial Differential Equations
Homework Problems

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Problem 1. From page 40 in the notes.

for the PDE: $u_t - c^2 u_{xx} = 0$, with initial condition $u(x,0) = g(x)$, show that the solution is given by the formula:

$$u(x,t) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} e^{-\frac{(x-\xi)^2}{4c^2 t}} g(\xi) d\xi \tag{1}$$

Solution. First compute the time derivative of u. Bringing $\frac{\partial}{\partial t}$ inside the integral gives

$$u_t = \int_{-\infty}^{\infty} \frac{\partial}{\partial t} \left[\frac{1}{2c\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{2}$$

Applying the product rule to the terms in brackets,

$$u_t = \int_{-\infty}^{\infty} \left[-\frac{1}{4c\sqrt{\pi}\, t^{3/2}} e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{1}{2c\sqrt{\pi t}} \cdot \frac{(x-\xi)^2}{4c^2 t^2} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{3}$$

Next compute the spatial derivative of u in the same way. Bringing $\frac{\partial}{\partial x}$ inside the integral gives

$$u_x = \int_{-\infty}^{\infty} \frac{\partial}{\partial x} \left[\frac{1}{2c\sqrt{\pi t}} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{4}$$

Since the prefactor does not depend on x, only the exponential is differentiated, so

$$u_x = \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \cdot \left(-\frac{x-\xi}{2c^2 t} \right) e^{-\frac{(x-\xi)^2}{4c^2 t}} g(\xi)\, d\xi. \tag{5}$$

Taking one more derivative of (5) with respect to x gives

$$u_{xx} = \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \frac{\partial}{\partial x} \left[-\frac{x-\xi}{2c^2 t} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{6}$$

Applying the product rule to the terms in brackets,

$$u_{xx} = \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \left[-\frac{1}{2c^2 t} e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{(x-\xi)^2}{4c^4 t^2} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{7}$$

Multiplying (7) by c^2 cancels some of the factors of c, so

$$c^2 u_{xx} = \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \left[-\frac{1}{2t} e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{(x-\xi)^2}{4c^2 t^2} e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \tag{8}$$

Substituting (3) and (8) into the PDE gives

$$\begin{aligned} u_t - c^2 u_{xx} = &\int_{-\infty}^{\infty} \left[-\frac{1}{4c\sqrt{\pi}\, t^{3/2}}\, e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{1}{2c\sqrt{\pi t}} \cdot \frac{(x-\xi)^2}{4c^2 t^2}\, e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi \\ &- \int_{-\infty}^{\infty} \frac{1}{2c\sqrt{\pi t}} \left[-\frac{1}{2t}\, e^{-\frac{(x-\xi)^2}{4c^2 t}} + \frac{(x-\xi)^2}{4c^2 t^2}\, e^{-\frac{(x-\xi)^2}{4c^2 t}} \right] g(\xi)\, d\xi. \end{aligned} \tag{9}$$

Since

$$\frac{1}{2c\sqrt{\pi t}} \cdot \frac{1}{2t} = \frac{1}{4c\sqrt{\pi}\, t^{3/2}}, \tag{10}$$

the two integrals in (9) have identical integrands, so everything cancels and

$$u_t - c^2 u_{xx} = 0. \tag{11}$$

□

Problem 2. From 3.3-1 in the book.

Let Ω be a smooth bounded domain in $\mathbb{R}^n$. For a C^2 solution of u of the wave equation $u_{tt} = c^2 \Delta u$, show that if the energy is defined as $E_\Omega(t) = \int_\Omega (u_t^2 + c^2 |\nabla u|^2) dx$, and either $u(x,t) = 0$ or $\frac{\partial u}{\partial n} = 0$ on $\partial\Omega$, then $E_\Omega(t)$ is constant in time.

Solution. To show that $E_\Omega(t)$ is constant, take its time derivative and set it equal to 0:

$$\frac{dE_\Omega}{dt} = \frac{d}{dt}\, \frac{1}{2} \int_\Omega \left(u_t^2 + c^2 |\nabla u|^2 \right) dx = 0. \tag{12}$$

Splitting this into two integrals,

$$\frac{d}{dt}\, \frac{1}{2} \int_\Omega u_t^2\, dx + \frac{d}{dt}\, \frac{1}{2} \int_\Omega c^2 |\nabla u|^2\, dx = 0. \tag{13}$$

Bringing $\frac{d}{dt}$ inside the first integral, u_t^2 becomes $2u_{tt}u_t$, which cancels the $\frac{1}{2}$ for that term:

$$\int_\Omega u_{tt} u_t\, dx + \frac{d}{dt}\, \frac{1}{2} \int_\Omega c^2 |\nabla u|^2\, dx = 0. \tag{14}$$

Bringing $\frac{d}{dt}$ inside the second integral and writing $|\nabla u|^2 = \sum_{i=1}^n u_{x_i}^2$, each $u_{x_i}^2$ becomes $2u_{x_i}u_{x_i t}$, which cancels the $\frac{1}{2}$ for that term. Since c is constant, we can also pull c^2 out of the integral:

$$\int_\Omega u_{tt} u_t\, dx + c^2 \int_\Omega \sum_{i=1}^n u_{x_i} u_{x_i t}\, dx = 0. \tag{15}$$

Next we show that the second integral in (15) is equivalent to the integral of $-u_t \Delta u$, by integrating by parts once in each x_i. Since $u_{x_i t} = (u_t)_{x_i}$, each term of the sum is $u_{x_i}(u_t)_{x_i}$. Integrating by parts in x_i moves the derivative off of u_t and onto u_{x_i}. By the divergence theorem, with n_i the i-th component of the outward normal,

$$\int_\Omega u_{x_i} (u_t)_{x_i}\, dx = \int_{\partial\Omega} u_t\, u_{x_i} n_i\, dS - \int_\Omega u_t\, u_{x_i x_i}\, dx. \tag{16}$$

Summing over i, and using $\sum_{i=1}^n u_{x_i} n_i = \nabla u \cdot n = \frac{\partial u}{\partial n}$ and $\sum_{i=1}^n u_{x_i x_i} = \Delta u$,

$$\int_\Omega \sum_{i=1}^n u_{x_i} u_{x_i t}\, dx = \int_{\partial\Omega} u_t\, \frac{\partial u}{\partial n}\, dS - \int_\Omega u_t \Delta u\, dx. \tag{17}$$

(This is Green's first identity with $f = u_t$ and $g = u$.) The boundary integral vanishes under either boundary condition. If $u = 0$ on $\partial\Omega$ for all t, then u does not change in time there, so $u_t = 0$ on $\partial\Omega$. If $\frac{\partial u}{\partial n} = 0$ on $\partial\Omega$, then that factor is zero directly. Therefore

$$\int_\Omega \sum_{i=1}^{n} u_{x_i} u_{x_i t}\, dx = -\int_\Omega u_t \Delta u\, dx. \tag{18}$$

Substituting (18) into (15),

$$\int_\Omega u_{tt} u_t\, dx - c^2 \int_\Omega u_t \Delta u\, dx = 0. \tag{19}$$

Combining the terms and factoring out u_t,

$$\frac{dE_\Omega}{dt} = \int_\Omega u_t \left(u_{tt} - c^2 \Delta u\right) dx = 0. \tag{20}$$

The second factor is the original PDE, which is 0. Since the change in energy is 0, $E_\Omega(t)$ is constant in time. □

Problem 3. From 3.3-2a in the book.

For the non-homogeneous wave equation $u_{tt} = \Delta u + f(x,t)$, show uniqueness of the solution u given the Dirichlet condition $u = g$ on $\partial\Omega$.

Proof. Assume we have two solutions u_1 and u_2 to the non-homogeneous wave equation, with the same boundary and initial conditions. Define $v = u_1 - u_2$. Then v satisfies the homogeneous wave equation:

$u_{1tt} = \Delta u_1 + f(x,t)$ in Ω
$u_{2tt} = \Delta u_2 + f(x,t)$ in Ω
$v_{tt} = \Delta v$ in Ω
$u_1 = g$ on $\partial\Omega$
$u_2 = g$ on $\partial\Omega$
$v = 0$ on $\partial\Omega$

By the same reasoning, we also have the initial conditions:

$v(x,0) = 0$ in Ω
$v_t(x,0) = 0$ in Ω

By the previous problem applied with $c = 1$, we know that the energy $E(t) = \int_\Omega (v_t^2 + |\nabla v|^2) dx$ is constant in time.

At $t = 0$, $v(x,0) = 0$ in Ω gives $\nabla v(x,0) = 0$, and $v_t(x,0) = 0$, so $E(0) = 0$. Since E is constant, $E(t) = 0$ for all t. The integrand $v_t^2 + |\nabla v|^2$ is continuous and nonnegative with zero integral, so $v_t = 0$ everywhere. Therefore $v(x,t) = v(x,0) = 0$, so $u_1 = u_2$. □

Problem 4. From 3.3-2b in the book.

For the non-homogeneous wave equation $u_{tt} = \Delta u + f(x,t)$, show uniqueness of the solution u given the Neumann condition $\frac{\partial u}{\partial n} = h$ on $\partial\Omega$.

Proof. Assume we have two solutions u_1 and u_2 to the non-homogeneous wave equation, with the same boundary and initial conditions. Define $v = u_1 - u_2$. Then v satisfies the homogeneous wave equation:

$u_{1tt} = \Delta u_1 + f(x,t)$ in Ω
$u_{2tt} = \Delta u_2 + f(x,t)$ in Ω
$v_{tt} = \Delta v$ in Ω

$\frac{\partial u_1}{\partial n} = h$ on $\partial\Omega$
$\frac{\partial u_2}{\partial n} = h$ on $\partial\Omega$
$\frac{\partial v}{\partial n} = 0$ on $\partial\Omega$
By the same reasoning, we also have the initial conditions:
$v(x,0) = 0$ in Ω
$v_t(x,0) = 0$ in Ω

By Problem 2 applied with $c = 1$, we know that the energy $E(t) = \int_\Omega (v_t^2 + |\nabla v|^2) dx$ is constant in time.

At $t = 0$, $v(x,0) = 0$ in Ω gives $\nabla v(x,0) = 0$, and $v_t(x,0) = 0$, so $E(0) = 0$. Since E is constant, $E(t) = 0$ for all t. The integrand $v_t^2 + |\nabla v|^2$ is continuous and nonnegative with zero integral, so $v_t = 0$ everywhere. Therefore $v(x,t) = v(x,0) = 0$, so $u_1 = u_2$. □

Problem 5. From 3.3-6 in the book.

Proof. We will use Green's first identity, which states that for a C^1 scalar field f and a C^2 scalar field g on $\overline{\Omega}$,

$$\int_\Omega (f\Delta g + \nabla f \cdot \nabla g)\, dx = \int_{\partial\Omega} f \frac{\partial g}{\partial n}\, dS. \tag{21}$$

Substituting u for both f and g in (21), and using $\nabla u \cdot \nabla u = |\nabla u|^2$,

$$\int_\Omega \left(u\Delta u + |\nabla u|^2\right) dx = \int_{\partial\Omega} u \frac{\partial u}{\partial n}\, dS. \tag{22}$$

If either $u(x,t) = 0$ or $\frac{\partial u}{\partial n} = 0$ on $\partial\Omega$, then the integrand on the right side of (22) vanishes, so that term is zero and

$$\int_\Omega \left(u\Delta u + |\nabla u|^2\right) dx = 0. \tag{23}$$

Rearranging (23),

$$\int_\Omega |\nabla u|^2\, dx = \int_\Omega -u\Delta u\, dx. \tag{24}$$

□